Bridging the Gap in Infrastructure Modeling & Visualization

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I. Introduction

Within the United States, there are more than 617,000 bridges each of which is paramount to ensuring the safe and reliable transit of goods, services, and individuals throughout the country. Despite their critical function, a significant proportion of the nation's bridges are in a state of disrepair, creating an issue of substantial public concern. However, this issue is not often recognized in part due to the lack of non-proprietary solutions for transparent exploration and visualization of the topic. Today's most recognized tool to explore bridge data, the "InfoBridge" web portal [1], lacks the clarity and flexibility expected of modern web applications.

II. Background & Problem Definition

This gap in existing technology creates the need for an open-source and user-friendly tool to better visualize, understand, and explore both existing national bridge data [2] and deterioration modeling approaches. The problem and how we intend to solve it is two-fold, composed of enabling a unified modeling and visualization framework for the U.S. bridge inventory systems that improve upon present solutions. This improvement will take the form of an interactive, user-friendly web application that offers simple access to bridge infrastructure data, visualizations, and models.

A. Visualization Survey

At present, the single freely available solution is the "InfoBridge" web portal [1]. The portal itself has data stored in a format not tailored to visualization design principles. Column names make data both inaccessible and confusing to expert and non-expert users alike and current geographic mapping lacks any interactivity or additional meaning beyond general location making the visualization uninformative. Performance and projection charts are visually unappealing and uninformative given the layout, size, and design of their associated sections thus limiting data exploration potential.

B. Modeling Survey

Three general categories broadly encompass the present body of academic work in bridge condition modeling: deterministic models, probabilistic models, and deep neural networks models. Deterministic models incorporate a set of defined relationships among features like time, age, region, etc. derived from human feature engineering and selection that would be used to model a given bridge component's need for replacement or repair based upon thresholds determined by engineers [3]. These initial approaches proved far too simplistic and inefficient for the task, leading to over replacement and poor resource allocation of components and labor. Gradually, their complexity increased through the adoption of linear regression models [4], polynomial regression models [5], and ensemble methods such as Random Forests [6]. However, their utility is limited by their rigid structure, susceptibility to multicollinearity, and reliance upon non-scaleable and often biased human feature engineering. Efforts have been made to remedy these shortcomings through variable selection methods such as LASSO and Step-wise Selection [7] as well as dimensionality reduction methods such as tensor decomposition [8]. Although effective in practice, this class of models is unable to quantify uncertainly with respect to inputs or predictions. This quality, which is key to risk assessment, is the primary reason for the creation of the probabilistic class of models. Probabilistic models were historically based upon failure and reliability distributions [9] parameterized by known sources of failure over a bridge's lifetime. However, due to the infrequent nature of bridge collapses and the inability of such an approach to dynamically adjust to temporally realized conditions, these approaches were replaced with more flexible stochastic process models. Presently the most utilized among these models are Markov chains [10] and as an extension of it, Bayesian Decision Networks [11] both of which permit the modeling of a bridge concerning its present known condition, a major improvement upon the static probabilistic models. However, the Markovian approach does not solve the issue of time dependence entirely as the process itself is memory-less, only the current state influences the next system state. This assumption directly contradicts the known dynamics of deep temporal dependencies throughout the deterioration process [12]. Furthermore, similar to early generation models there is still significant subjective feature engineering and distributional approximation taking place in the construction of the transition probability matrices of these decision networks and Markov processes [13]. AI models directly addresses this issue as well as the necessity of expert feature engineering which plagues both deterministic and probabilistic models. This is most commonly achieved through the utilization of deep artificial neural networks. These computational models utilize several processing layers to learn and create a hierarchy of abstraction to best represent the training set [14]. A wide variety of neural network architectures have been proposed from dense models [15] to convolutional networks [16] that more effectively engineer features through up-sampling of the feature space. Lesser known, but still relevant alternative hybrid AI methods include clustering approaches through KNN [17] and K-means [18] [6] as a feature engineering or prepossessing step to more effectively apply subsequent probabilistic modeling approaches. Furthermore, AI methods are far less transparent especially to non-expert users compared to the prior two classes of models, and still do not provide a complete solution to the temporal aspect of the problem. Despite the substantial body of work that composes present modeling approaches for bridge deterioration and forecasting, there is still significant room for improvement. The most prevalent short-coming which can be attributed to all contemporary modeling approaches is the lack of data integration across some of the most crucial known causes of bridge deterioration such as ambient temperature [19], climate [20], traffic, and accident data [21], and utilizing only a fraction of the national data available. Collectively, each of these historical shortcomings has motivated and shaped the development of our proposed improved methodology of bridge deterioration modeling.

III. **Methodology**

The defining differences in our project's methodology and the existing body of work within the field come down to four primary areas of innovations we plan to implement that collectively vastly improve the state of the art. Firstly, our new approach to infrastructure design and integration allows for access by both desktop and mobile users with significant speed ups created by our back-end infrastructure. Secondly, our expanded data aggregation, consolidation, and cleaning efforts borrow from the best of the existing literature base while expanding to promising new features such as weather & traffic. Similarly, our modeling approach also benefits from this more expansive feature space, and when combined with more modern and refined model architectures, ultimately allows more comprehensive models and more accurate predictions. Finally, our new visualization methodology adds a new layer of potential for data exploration and presentation via our central focus on interactivity. A further discussion of our project's innovations in methodology are discussed in the following sections.

A. **Infrastructure Methodology**

Robust and flexible infrastructure is key to the operation and accessibility of our tool. With the project's emphasis on easy data exploration and accessibility, our team recognized the importance of creating an open-source and fast tool. As such, we decided to use Python's Flask web framework to host the application's web interface, a MySQL database for data storage, and Tableau for our visualizations. We will use Amazon Web Services (AWS) to bring these pieces together, with an Elastic Compute Cloud (EC2) instance for application hosting, a Relational Database Service (RDS) MySQL instance to serve the database, and a Simple Storage Service (S3) bucket to store files. **Figure 1** details the infrastructure. Brought into the EC2 instance through an HTTP request, calls will be made to the MySQL database and the S3 bucket to retrieve data and files. Data will also be tunneled through Tableau Online, which will produce our page's embedded visualizations. Given our project's goal is to fill the present void in user-friendly and open-source tools surrounding this data, the infrastructure was uniquely selected to support quick development, flexibility, and ease-of-use. The Flask framework offers simplicity, giving us detailed control over how our application is presented. With Flask serving HTML, we have the capabilities of HTML, CSS, JavaScript, etc. to fully customize the interface. This leads to innovation that goes beyond the existing tools: with Flask we can make both desktop and mobile layouts, integrate JavaScript and D3, and provide a lightweight tool that works on all browsers. We decided on a MySQL database as it is fast, reliable, and open-source. Packages such as SQL Alchemy make it simple to write/read data directly via Python scripts, which is necessary for our Flask-based application. MySQL also easily connects to Tableau Online, the visualization tool we will use to present our data. At our infrastructure's core, AWS grants us configurability and near-perfect up-time, allowing us to focus on features, visuals, and models. By hosting on EC2, we do not need to manage a physical server. By using an MySQL RDS instance, we don't need to do configuration beyond adding data objects. Overall, the major benefit of using these AWS services is that we can quickly scale depending on our needs. If we need more than 20GB of storage in our RDS instance, we can increase our storage without downtime. Likewise, we can increase the computing power of our application as requirements increase. As projected in our proposal, AWS costs still remain at $0 due to the AWS credits generously allocated for the course.

B. Data Preparation Methodology

The first step in the data workflow was the aggregation of several data sets we utilized for modeling and visualization purposes later in the report. The data utilized in this project has been collected from three primary sources. Firstly, all data concerning the location, status, and condition of bridges from years 1993-2020 was pulled from the National Bridge Inventory database [2]. Monthly traffic volume and accident data was collected from 2002-2020 via the FHA traffic volumes database [22], and was linked to corresponding bridges via route and regional encoding information. Finally, weather data was collected from the NOAA monthly weather database [23] utilizing the report stations' latitudes and longitudes to match with bridges. Despite our data sources originating from government endpoints, there was an immense amount of cleaning and data pre-processing required prior to doing any modeling or visualization work. Since clean and accurate data is the cornerstone to any thorough analysis, we spent a significant amount of time preparing and cleaning the dataset. First was the consolidation and merging of 27 years of independently stored bridge data. This task was further complicated due to the inconsistent conventions used in data entry from year to year which had to be manually adjusted. Inconsistent data conventions across years added complications, leading us to make manual adjustments. Likewise, there were instances of duplicated bridge records that had to be reconciled. Additionally, since most of the data was coded according to internal numeric NBI conventions, we converted this data to human-readable text. Due to our focus on the 48 contiguous U.S. states, bridges presiding in territories, Alaska, and Hawaii were removed. After cleaning the bridge data set, we began matching each bridge with a corresponding NOAA weather station based on the Haversin distance formula which is used to calculate distances between given latitude and longitude endpoints [24]. By pairing each bridge with the closest station, we save storage space and provide more accurate information about each bridge's unique exposure to weather and climate. Next, we began converting the FHA traffic volumes data set into matching fields for each bridge by year. For cleaning, we only had to identify and remove outliers. The cleaning and merging of the traffic data set on each bridges' respective route was straight forward, only requiring outlier identification and removal. Overall, these unexpected difficulties led to changes within our project timeline. Initially, we planned to quickly create a sample data set for use while the full data set was being cleaned. However, it quickly became clear that preparing a subset of data would be just as difficult as preparing the entire data set. Thus, the timeline for data operations was expanded a week from our initial estimation. After cleaning operations were complete, the data was made available to the visualization and modeling teams via our MySQL database. Here, various virtual tables and indexes were constructed to permit fast querying for common requests and queries. We do not regret re-prioritization as a clean data set provides the foundation for any future visualization and modeling so the additional time was well allocated.

C. Modeling Methodology

After completion of data aggregation and cleaning, we began the data pre-processing necessary for modeling. To optimize our ability to properly model future bridge conditions, it became clear that it makes the most sense in terms of accuracy to divide the U.S. into nine geographic regions instead of trying fit one model for all U.S. bridges. Then, after filtering all the bridges for a given geographical region, we would then fit a model on just that subset, repeating this process on the remaining 8 regions. This has much to due with the inherently large differences in weather, climate, and land-forms that encompass the diverse geography of the U.S. By creating regional subsets, we were able to better isolate important factors specific to the bridges within each region which is further supported by existing literature [25]. For each regional subset, we will conduct feature selection and dimensionality reduction experimenting with different selection criteria and methods. As the feature space for this problem is vast, this took a significant proportion of the modeling team's time. Many of the features required significant pre-processing due to high cardinally among some predictors. Decisions were made on their inclusion or removal in order to reduce the effect of the curse of dimensionality. Additionally, several features contained a large proportion of null values which would be uninformative and confounding to modeling efforts, and were thus removed. After pre-possessing was completed, we tested a variety of models for use in the prediction task of future bridge condition. These experiments and results are described in further detail in subsequent sections. After results and predictions were generated form the production models resulting from our experiments and evaluation phase, we utilized a test bed to verify the logical consistency of our results. For example, it would be impossible for a bridge to improve in condition over time without being scheduled for repairs within that time frame. Similar test cases like the previous statement were used to ensure that the modeling results were not removed from reality as well as a sanity check in the testing process. The results of the test bed were ultimately satisfactory with only a negligible number of edge cases that needed to be resolved.

D. Visualization Methodology

For our visualizations we utilized Tableau [26], hosted on Tableau Public, to embed dashboards within our application. We had two main visualizations. The first, a map chart of the contiguous U.S., provided detailed information about every bridge at its specific GPS location. The second visualization provided a way to interact with the model results and see how these results varied across different dimensions. Interactivity was achieved through the use of tool-tips and filters, allowing the user to make various selections and display detailed bridge records, statistics, and other relevant information such as historical weather trends and traffic patterns. The visualization team's first focus was to parse the data stored in the database. As each bridge's structure name and latitude/longitude coordinates were used as the key indexes, the parsing process was important in verifying the availability of structure names and formatting the coordinates for use with Tableau. To boost performance and minimize data processing, the data team helped reformat and rearrange key indexes for improved querying. To create the first dashboard, two Tableau sheets along with a detailed tool-tip were utilized. The first sheet was a map of the U.S. with individual bridges displayed by using their latitude and longitude coordinates; each bridge was represented by a dot colored by traffic direction. By hovering over this dot, a tool-tip showing essential bridge information, including the model results, was displayed. To decrease latency, bridges were grouped by region and state. The second sheet in the dashboard displayed aggregated traffic information and the estimated volatility's of bridges in the selected states and regions. High-level metrics about the selected bridges were also displayed, with the number of bridges in each class and each bridge's structure type broken down by the model prediction. Paramters were also available to allow dynamic-selection of different model results.

One of the big challenges was visualizing datasets of large volume and high level of detail. In our initial attempts at joining the tables in Tableau, we found the large data volume made the visualizations very slow. To work around this issue, we joined the tables together in the database, aggregating traffic and weather data for each bridge; this created a much better user experience.

In order to improve upon the lack of transparency in intermediate modeling results of bridge deterioration modeling and approaches utilized within existing web applications, we created a standalone interactive dashboard to display all relevant modeling results and performance. Utilizing the modeling output plots of each regional models' respective metrics such as loss, accuracy, and other model specific information generated during the cross-validation process, we created an organized and interactive experience for users to better understand both our intermediate and ultimate modeling process. In order to display the information in an effective manner within the Flask application, we utilized JavaScript and D3 to create an interactive regional map of the United States which updates modeling results and images displayed on the page upon a regional selection from the provided drop-down box with the information relevant for that specific region. This new, novel, and interactive dashboard represents a direct improvement in both transparency and usability to existing web applications.

IV. Plan of Activities & Work Distribution

In our proposal, we noted that our team is broken into the data, modeling, and visualization sub-teams: Ben and Jacob of the data team being responsible for cleaning/integrating data and database maintenance, Evan and Mernegar of the modeling team being responsible for modeling and back-end optimization, and Brian and Travis of the visualization team being responsible for creating visualizations. We also noted we'd measure progress in deliverable checkpoints at 3 and 6 weeks. This sub-team breakdown remained throughout the project, with each of the tasks charged to specific group members (noted in previous reports) being completed by the assigned individual within the necessary time constraints. All self-imposed internal project deadlines and checkpoints were met or exceeded.

In terms of team member effort, Jacob and Evan led the main elements of the project, contributing substantial time and effort. They also took a lead on their respective sub-teams. Travis was a strong contributor and leader of the visualization sub-team, focusing primarily on all the components of that work. Brian made fair contributions to the project's visualization-component. Mernegar focused primarily on the modeling aspect of the project. Ben made contributions to the data sub-team.

V. Modeling Experiments & Evaluation

The vast majority of experimentation for this report is centered on the various modeling approaches utilized in developing our production bridge deterioration models. In the model selection phase, we began by using our existing literature survey to narrow the space to seven potential models: random forest, AdaBoost, naive Bayes, logistic regression, artificial neural network, support vector machines, and Markov chain models. Model comparisons were later made by conducting 5-fold cross-validation on regional data subsets using each of the models (excluding Markov chains which will be discussed separately as it is a purely probabilistic model not applicable to cross-validation). During this cross-validation phase, several things quickly became clear. First, due to the problem's nature and the size of the training folds, the long run-time required to fit the model made the support vector machine especially infeasible. For this reason, the model was excluded from subsequent analysis. Similarly, the traditional Markov chain approach faced similar issues caused by the dataset's underlying high-dimensional structure which makes the model's state-space intractably large. With this in mind, the traditional Markov approach for joint state prediction across the entire training set was abandoned. Despite this initial setback however, Markov modeling of the individual features and their distribution of future states was still found to be insightful and useful from a modeling perspective. By generating these individual estimates of future state distributions for all temporally relevant features, we were able to create meaningful forward-looking volatility estimates for each bridge. Furthermore, these state distributions permitted the simulation and subsequent extrapolation of trained models to predict bridge conditions beyond the most recent year. To our knowledge, this is the first time such an approach has been applied to this problem, representing a significant and novel innovation to our modeling approach. With the field narrowed to only five models, we finished the cross-validation process and observed a significant, consistent difference in the categorical accuracy performance across all folds, with logistic regression and naive Bayes showing similar and consistent under-performance to the higher performing models of random forests, AdaBoost, and neural networks. This segmentation of models is very insightful as it demonstrates the complexity of the required decision boundary. Although flexible enough for many difficult modeling tasks, naive Bayes and logistic regression are not capable of modeling deeply-complex decision boundaries due to their limiting underlying structure. However, ensemble methods like AdaBoost and random forests rely upon the use of several weak learners, creating the ability to model far more complex boundaries. Similarly, artificial neural networks also vastly expand the ability to fit complex boundaries when compared to naive Bayes and logistic regression. While it was welcome news to see the strong performance by AdaBoost, random forests, and the neural network modeling approaches, the stronger performance of these more complex models came at the cost of decreased interpretability. Despite this trade-off, we decided that these three models were the best choice for the modeling task moving forward. For each region, we conducted further hyper-parameter search and reported the best resulting model of each. We found that regardless of the region the random forest model, albeit with different hyper-parameters, universally was the best. For the random forest grid search, the hyper-parameters of interest were number of estimators and maximum depth of each tree. For the AdaBoost models, number of estimators and learning rate were evaluated. Finally, for the neural network models, ten different architectures were tested with varying layer sizes, numbers, and activations. The project's modeling page showcases each of the individual parameters that were tested and the respective results. Generally we saw similar hyper-parameter behavior/performance across all regions. With respect to random forests, most models performed best with increasing maximum depth and number of estimators. Similarly, AdaBoost also benefited from an increasing number of estimators, but was more sensitive to high learning rates. The neural network models universally performed best with 4 layers (beyond or below which performance was significantly degraded) without any intermediate dropout applied, and with softmax as opposed to sigmoid output activation. Random forest models displayed the most superior performance and are generally recommended as the best modeling approach. Our resulting random forest models outperform the present state-of-the-art modeling approaches. This was evaluated on the grounds of improved generalization across all contiguous forty-eight states (which to our knowledge hasn't been attempted to date) and on our achievement of a categorical accuracy rate of 99.24% comparable to the leading neural network approach by Heng Liu and Yunfeng Zhang [16] that achieved 85% categorical prediction accuracy.

VI. Visualization Experiments & Evaluation

At the onset of this project, we realized that there were few, if any, modern and easy-to-use tools for exploring bridge infrastructure data interactively. As such, our goal in this project's pursuit was to exceed the capabilities and functionality of existing tools, the primary one being the Federal Highway Administration's InfoBridge portal, by creating a new application that serves bridge data in clear, interactive, and functional ways. As we've stated in other sections of this report, finding and making use of supporting tools that are both feature-rich and popular—such Tableau Public and Python Flask—was the first step in giving us an edge over the existing options. By then thoughtfully developing interesting and highly-interactive visualizations, we were able to develop an application that we believe is better than existing options. However, we realized that our team's personal evaluation of our new application does not necessarily reflect how a more widespread audience may evaluate our tool compared the prevailing options. Thus, it was important for us to develop a system of experiments, and methods to evaluate them, to best understand the success of our project.

As such, to evaluate the success of our project implementation and how it compared to the existing solution, we created a experiment to gather feedback. This experiment, a survey in a Google Form, was distributed to friends of the team members. It contained questions about our visualizations, model output, and overall application, asking the respondents to then answer the same questions about the existing InfoBridge portal. Each of the five questions were evaluated on an 5-point Likert scale. An example can be seen in **Figure 3**. The questions were as follows:

- How would you rate the application overall?
- How would you rate the overall usability and ease-of-use of the application?
- How would you rate the responsiveness and visual appearance of the application?
- Do you clearly understand the data that you are viewing?
- How would you rate the functionality of the application?

In total, this survey received 10 responses. From this data, our application received an average rating of 4.5 out of 5, scoring the highest on overall functionality and ease-of-use. The InfoBridge application received an average of 2.5, scoring well on data clarity but lower on the other metrics. Although the rigor of the survey could be improved, and the results are potentially biased due to the respondents not being a random sampling, we believe that it still demonstrates positive reception of our application overall. More specifically, it also signals that our application is objectively considered easier to use and more functional than the existing InfoBridge tool. As such, despite the survey respondents potentially being biased due to their connection's with team members, we believe this experiment provided us with valuable insight into our tool's overall perception and usefulness.

VII. Concluding Thoughts

Overall, based on the empirically-measurable performance advantages of our models, our use of modern visualization tools and infrastructure, and the surveys we conducted to better understand the opinion of our visualization approaches, it is clear that we were successful in achieving our project's stated goals. We successfully created bridge condition models that out-perform the industry's current approaches, we built a primarily open-source application that makes infrastructure widely available and easily accessible for all, and we built clear, robust, and informative visualizations that make it simple to understand the current state of bridges in the U.S. Looking beyond present accomplishments, it is apparent that future research and experimentation with greater resources would greatly benefit both our application and field as a whole. Barring present constraints, we ideally would've spent additional time on preprocessing and feature engineering giving each region a more fully individualized treatment with accompanying expanded hyper-parameter search consistent with the treatment of the Southeast region (our primary focus). All in all, this application will serve as a valuable tool for anyone looking to view and understand bridge infrastructure data in a efficient, functional, and visually-appealing way. As it is an open-source application, we hope to see further contributions to our project leading to a better and more informative tool as time goes on.

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APPENDIX I

Fig. 1. Project Modeling Diagram

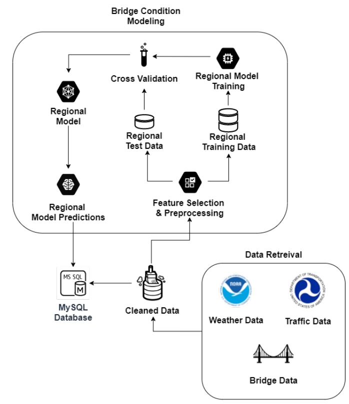

Fig. 2. Project Infrastructure Diagram

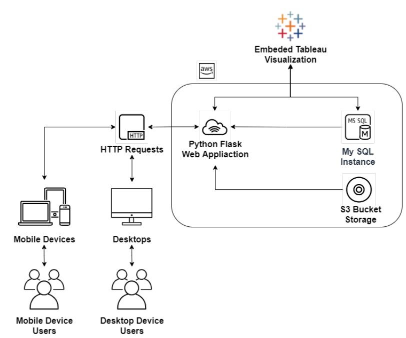

Fig. 3. Example Survey Question